Towards Multi-Permeability Cores Through Composition-Controlled Pellet Extrusion

1 Background

Reducing the volume of inductors and transformers is an important route to increasing power density in power converters [1]. Their soft magnetic cores guide magnetic flux and influence electromagnetic performance [2]. The magnetic flux distribution within a core depends on the geometry, material properties and operating conditions. Nonuniform flux density can lead to uneven core losses and localised heating [3]. In some inductor designs, regions near the conductor approach magnetic saturation while material farther away remains underutilised [4]. Near saturation, further increases in magnetising field produce relatively little additional flux density as the material's incremental permeability decreases. This can reduce the effective inductance as the current increases, limiting the utilisation of the core [2, 5]. These limitations motivate designs that distribute magnetic flux more effectively.

One approach is to optimise the core geometry and winding arrangement. For example, Cui et al. designed an inductor with an improved flux distribution by shaping the core and distributing the winding turns [6]. A complementary approach is to vary permeability spatially within the core to tailor the flux distribution [4, 7]. Wang et al. experimentally demonstrated that a three-permeability inductor achieved higher inductance than a single-permeability counterpart and improved the efficiency of the tested high-frequency DC/DC converter [4]. Liu et al. reported that a three-permeability toroidal core achieved the same inductance at a smaller volume than a single-permeability core [7]. These results demonstrate the potential benefits of spatial permeability variation for specific designs, although the benefits depend on the core geometry and operating conditions. Geometry and permeability are not mutually exclusive design variables, and optimising both may offer further benefits [8].

Several manufacturing approaches have been investigated to realise spatially varying permeability. Lu et al. produced discrete radial permeability regions by compacting individual rings at different pressures [9]. Continuous radial permeability variation has also been demonstrated by imposing a temperature gradient during sintering [10]. These approaches demonstrate the feasibility of creating multi-permeability cores, but the reported implementations rely on specific core geometries and manufacturing arrangements.

Additive manufacturing offers an alternative for producing customised geometries and positioning different material formulations within a component. Several AM processes have been investigated for soft-magnetic materials, for example powder-bed fusion, binder jetting and material extrusion [11, 12]. The suitability of each process depends on the feedstock, required magnetic properties and post-processing conditions. Material extrusion is particularly relevant for thermoplastic polymers and composites because it offers relatively low initial investment, simple operation and the capacity for simultaneous use of multiple materials [13].

Liu et al. demonstrated that material extrusion can be used to manufacture multi-permeability cores. They employed a multi-extruder paste-based 3D printer to fabricate a toroidal core with three distinct permeability regions [7]. Their ferrite paste required sintering above 900 °C, whereas their powdered-iron paste could be cured below 250 °C. This demonstrates the feasibility of additive manufacturing for multi-permeability cores, while illustrating the importance of feedstock formulation and post-processing requirements.

Within material extrusion, the choice of feedstock involves a trade-off between magnetic performance and manufacturability. Producing filament-based composites with high magnetic-particle loading while retaining suitable processability is challenging [14]. Cañada et al. report that filament-based soft-magnetic cores in the studies they reviewed generally exhibited relative permeabilities below 10. Paste-based approaches can achieve much higher permeabilities, but their rheology can complicate dimensional accuracy, and some formulations require

high-temperature sintering. In contrast, pellet-based feedstock can provide moderate permeability without high-temperature post-processing. Cañada et al. demonstrated nylon-12/FeSiAl composite cores with a relative permeability of approximately 42 using pellet-based material extrusion [15]. Since these results come from different materials and studies, they do not constitute a controlled comparison of the three feedstock types. Nevertheless, pellet extrusion provides a promising starting point for this work because it has already demonstrated useful soft-magnetic properties without high-temperature post-processing. Additionally, Cañada et al. reported variation in permeability for printed nylon-12/FeSiAl composites between different frequencies. Therefore, the permeability of the printed composites should be characterised at defined frequencies to establish whether changes in pellet blend ratio.

The composition of a polymer-based magnetic composite influences its magnetic properties. Hu et al. experimentally investigated the permeability of polymer composites with different magnetic inclusion contents, although their matrix differed from the nylon-12 composite considered here [16]. Huber et al. demonstrated variable-composition printing by blending NdFeB-filled PA12 filament with unfilled PA12 filament. Their approach controlled the remanent magnetisation distribution of hard-magnetic permanent magnets to produce a predefined stray field, rather than the effective permeability of a soft-magnetic core [17]. This demonstrates that magnetic composition can be varied during material extrusion, but does not establish how pellet blend ratio affects the permeability of a soft-magnetic composite.

Composition-controlled pellet extrusion has also been investigated in non-magnetic material systems. Brackett et al. studied in-situ blending of neat and carbon-fibre-filled ABS pellets and showed that transitions between compositions occupy a finite distance and depend on the direction of the material change [18]. This indicates that changes in the input blend ratio may not translate immediately into corresponding changes in the extruded material. However, the material system and large-format extrusion process they investigated differ from the present application. Elsersawy et al. found that dedicated mixing elements improved the uniformity of blended and coloured PLA compared with a standard Mahor V4 pellet extruder without a mixing element [19]. Although this demonstrates the importance of mixing configuration, colour uniformity does not establish uniform magnetic filler distribution or reproducible permeability. It therefore remains unclear whether a standard pellet extruder provides sufficient mixing to produce predictable and repeatable magnetic properties when blending magnetic-filled and unfilled polymer pellets.

2 Problem Statement

The practical challenge is to determine whether blending magnetic-filled and unfilled polymer pellets using a standard pellet extruder provides a reliable means of controlling the effective permeability of the printed material. The nominal blend ratio may not correspond directly to the local composition of the extruded material, while residual material can complicate transitions between different blends during printing. This work will therefore characterise the blend-ratio–permeability relationship, assess mixing-related variation, and evaluate whether a controlled pause–purge–resume procedure can be used to manufacture a core with three distinct permeability regions.

2.1 Research Question

To address the discovered problems, the work will investigate the following research questions:

> *Can varying the blend ratio of magnetic-filled and unfilled polymer pellets produce reproducible changes in the permeability of a soft-magnetic composite manufactured using a commercial pellet extruder?*

How closely does the measured bulk FeSiAl content match the content expected from the nominal blend ratio, and how are bulk-composition and local filler-distribution variations associated with permeability variability?

Can discrete permeability regions be fabricated by sequentially printing pre-blended pellet batches, using controlled purging to achieve distinct and reproducible material compositions?

2.2 Requirements

To address the research question and capture the overarching scientific plan, the following requirements have been defined:

- Establish a repeatable pellet-extrusion process for producing test specimens from the selected magnetic-filled and unfilled polymer pellets.
- Quantify bulk filler-content variation and local FeSiAl distribution in a representative 50:50 pellet blend across selected extrusion positions and repeat runs.
- Determine the effective permeability and between-specimen variability for five magnetic-filled-to-unfilled pellet mass ratios (0:100, 25:75, 50:50, 75:25 and 100:0), using three independently printed specimens per ratio.
- Establish a repeatable pause–purge–resume procedure that limits residual material from the previous pellet batch when switching blend ratios.
- Evaluate the feasibility of fabricating a core with three distinct permeability regions by sequentially printing pre-blended pellet batches.
- Characterise the three-region core's inductance and, where resources permit, saturation behaviour and core losses, and compare its performance with a uniform-permeability reference core of identical geometry.

2.3 Methodology

To fulfill the setout requirements, the following methodology has to be used:

- Conduct a targeted review of commercially available pellet extruders and compatible polymer feedstocks. Create a evaluation matrix, setting out project criterion with their specific weight. Select the highest rated path of solutions. Develop and assemble the selected manufacturing process. Investigate appropriate printing parameters.
- Manufacture a homogeneity specimen at 50:50 blend ratio. Use Thermogravimetric Analysis (TGA) to estimate bulk filler content, calibrated to the known filler content. Scanning Electron Microscopy paired with Energy-Dispersive X-ray Spectroscopy (SEM–EDS) mapping of cross-sections to quantify local FeSiAl distribution. Samples from independent extrusion runs and different specimen locations will be analysed to assess compositional variation, mixing homogeneity and run-to-run reproducibility.
- Choose and design a fixed rectangular toroid specimen geometry. Using the developed manufacturing process, fabricate three specimen at each specific (0:100, 25:75, 50:50, 75:25, 100:0) magnetic-filled-to-unfilled pellet mass ratios. Ensure that the printing parameters are constant. Wind the same number of turns of isulated copper wire around each specimen, ensuring consistent winding arrangement. Measure the inductance and it's between-specimen variability using an impeadance analyzer across a specific frequency range. Calculate the effective permeability.

- To determine the required purge amount, a print will be paused after reaching steady-state extrusion with a 0:100 blend, and the remaining pellets will be removed from the hopper before loading the 100:0 blend. The extruder will then be purged into separately collected samples at defined cumulative extruded masses. TGA will be used to estimate the composition of each sample, using known blend ratios to establish a calibration. The composition will be plotted against the cumulative purge mass to determine how much material must be purged before the new blend reaches a stable composition. The procedure will be repeated to additionally for the transition from 100:0 to 0:100 ratio.
- Design a toroidal core with a rectangular cross-section and three regions (100:0, 75:25 and 50:50) arranged to accommodate sequential material changes. Assign each region a blend ratio based on the experimentally characterised permeability values. Manufacture the core using the established pause–purge–resume procedure. Visually assess whether the intended regions have distinct compositions.
- Wind the same amount of turns of isulated copper wire around the manufactured core. Measure the inductance of the multi-permeability specimen and uniform permeability specimen with a impeadance analyzer across a specific frequency range. Remove the winding. Wind an excitation coil and a separate sensing coil around the core. Drive the excitation coil with a controlled AC source. Measure the excitation current and sensing coil voltage. Calculate the magnetic field strenght and flux density. To measure the core losses, adjust the source so the core reaches the peak flux density. Record the B-H loop. Calculatethe energy loss from its enclosed area. Repeat the procedures for uniform permeability specimen. Compare the resuls.

2.4 Delimitations

The project shall use the available commercial pellet extruder and the selected magnetic-filled and unfilled polymer feedstocks as the baseline. The project will not investigate the effect density and porosity has on the effective permeability. The development of an automated, continuously adjustable multi-material feeding system is outside the initial scope.

References

[1] Charles R. Sullivan, Di Yao, Garet Gamache, Alexander Latham, and Jizheng Qiu. Passive Component Technologies for Advanced Power Conversion Enabled by Wide-Band-Gap Power Devices. 41(8):315–330.

[2] Vencislav Cekov Valchev. *Inductors and Transformers for Power Electronics*. CRC Press.

[3] Edris Agheb and Hans Kristian Høidalen. Modification of empirical core loss calculation methods including flux distribution. 7(5):381–390.

[4] Laili Wang, Zhiyuan Hu, Yan-Fei Liu, Yunqing Pei, and Xu Yang. Multipermeability Inductors for Increasing the Inductance and Improving the Efficiency of High-Frequency DC/DC Converters. 28(9):4402–4413.

[5] Kazuhiro Umetani. Improvement of saturation property of iron powder core by flux homogenizing structure. 8(6):640–648.

[6] Han Cui, Khai D. T. Ngo, Jim Moss, Michele Hui Fern Lim, and Ernesto Rey. Inductor Geometry With Improved Energy Density. 29(10):5446–5453.

[7] Lanbing Liu, Chao Ding, Shengchang Lu, Ting Ge, Yi Yan, Yunhui Mei, Khai D. T. Ngo, and Guo-Quan Lu. Design and Additive Manufacturing of Multipermeability Magnetic Cores. 54(4):3541–3547.

[8] Vinicius Cabral Do Nascimento, Seung-Ryul Moon, Kevin Byerly, Scott D. Sudhoff, and Paul R. Ohodnicki. Multiobjective Optimization Paradigm for Toroidal Inductors With Spatially Tuned Permeability. 36(3):2510–2521.

[9] Panpan Lu, Rongzhi Zhao, and Yunhui Mei. Design and Preparation of Low-temperature Curable Multi-permeability Inductor with Low Power Loss for High Frequency Applications. pages 1–1.

[10] Junchang Liu, Yunhui Mei, Shengchang Lu, Xin Li, and Guo-Quan Lu. Continuously Variable Multi-Permeability Inductor for Improving the Efficiency of High-Frequency DC–DC Converter. 35(1):826–834.

[11] Feng-Hui Wang, Cai-Yin You, Na Tian, He-Guang Liu, Jing Zhang, and Xiao-Pei Zhu. 3D printing of soft magnetic materials: From printing to applications. 990:174486.

[12] Xiangxia Wei, Ming-Liang Jin, Haiqiang Yang, Xiao-Xiong Wang, Yun-Ze Long, and Zhangwei Chen. Advances in 3D printing of magnetic materials: Fabrication, properties, and their applications. 11(5):665–701.

[13] Mahrukh Sadaf, Mario Bragaglia, Lidija Slemenik Perše, and Francesca Nanni. Advancements in Metal Additive Manufacturing: A Comprehensive Review of Material Extrusion with Highly Filled Polymers. 8(1):14.

[14] Anthony P. Taylor, Camilo Velez Cuervo, David P. Arnold, and Luis Fernando Velasquez-Garcia. Fully 3D-Printed, Monolithic, Mini Magnetic Actuators for Low-Cost, Compact Systems. 28(3):481–493.

[15] Jorge Cañada, Steven W. Wright, Michail E. Kiziroglou, Eric M. Yeatman, and Luis F. Velásquez-García. Magnetic Flux Guides by Material Extrusion. 13(12):e17310.

[16] Xiaocheng Hu, Zipeng Liang, Qiyang Li, Sideng Hu, and Shaoxing Qu. Experimental study on the magnetic permeability of inclusion filled soft polymeric composite for soft-core transformer applications. 106:107430.

[17] Christian Huber, Claas Abert, Florian Bruckner, Martin Groenefeld, Stephan Schuschnigg, Iulian Teliban, Christoph Vogler, Gregor Wautischer, Roman Windl, and Dieter Suess. 3D Printing of Polymer-Bonded Rare-Earth Magnets With a Variable Magnetic Compound Fraction for a Predefined Stray Field. 7(1):9419.

[18] James Brackett, Yongzhe Yan, Dakota Cauthen, Vidya Kishore, John Lindahl, Tyler Smith, Zeke Sudbury, Haibin Ning, Vlastimil Kunc, and Chad Duty. Characterizing material transitions in large-scale Additive Manufacturing. 38:101750.

[19] Rawan Elsersawy, Andrew Rowe, Chance Smith, Dawson Halstead, Erik DeRosier, Mainul Hossain, and Mohammad Abu Hasan Khondoker. Design and performance assessment of custom static intermixers in extrusion 3D printing using machine learning–driven image analysis. 15(1):36557.